

CURRICULUM VITAE

Lily Larkins

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***Professional Interest:** I am particularly interested in translating complex astrophysical research into accessible stories that inspire curiosity and understanding of the universe. My background in exoplanet research and astrophysics, combined with experience leading public stargazing programs and STEM education, motivates my work in connecting cutting-edge scientific discoveries with broad audiences through writing, media, and public programming.*

EDUCATION

University of California, Santa Cruz - B.S. in Astrophysics GPA: 3.37

RESEARCH EXPERIENCE

Undergraduate Research Assistant May 2024 - Current
University of California, Santa Cruz | Advisor: Dr. Artem Aguichine

- ❖ Developed and tested Equations of State (EOS) for molten mantle composition.
- ❖ Implemented new EOS into the existing Fortran interior structure model to predict new planetary radii.
- ❖ Draft in prep for publication in a peer reviewed journal.

Undergraduate Research Assistant January 2024 - June 2024
University of California, Santa Cruz | Advisor: Dr. Francis Nimmo

- ❖ Digitized and scaled lunar magnetic field data from Lunokhod 2 rover mission.
- ❖ Extrapolated measurements with Python to match lunar coordinates, contributing to comparative lunar studies.

Undergraduate Observational Experience September 2024 - December 2024
University of California, Santa Cruz | Professor: Rebecca Jensen-Clem

- ❖ Remote observing experience on the Nickel telescope located at Lick Observatory on Mt. Hamilton.
- ❖ In-person observing experience on the Shane telescope, also located at Lick Observatory.

- ❖ Gained experience using the CCD Camera on the Nickel and the spectrograph on the Shane.

TEACHING AND OUTREACH EXPERIENCE

Wyoming Stargazing - Jackson, Wyoming
Stargazing Leader

August 2025 - Current

- ❖ Lead stargazing programs in Grand Teton National Park, engaging diverse audiences in hands-on observational astronomy experiences.
 - ❖ Provide detailed explanations of celestial objects, sky navigation techniques, and the use of astronomical equipment.
 - ❖ Set up, calibrated, and operated a 22-inch Dobsonian telescope to offer participants close-up views of planets, stars, nebulae, and other astronomical phenomena.
 - ❖ Facilitate educational discussions tailored to different age groups and levels of astronomy knowledge.
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Summit Innovations School - Jackson, Wyoming
STEM Instructor, Avid Tutor, Substitute Teacher

September 2025 - Current

- ❖ Provide individualized academic support and tutoring in mathematics, physics, and chemistry to high school students attending a continuation school focused on credit recovery and alternative education.
 - ❖ Work closely with students from diverse backgrounds, including those navigating academic and personal challenges.
 - ❖ Adapted teaching methods to accommodate a variety of learning styles, with an emphasis on patience, equity, and building student confidence.
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Colter Elementary School - Jackson, Wyoming
After the Bell Club Leader - Stargazing Club

September 2025 - November 2025

- ❖ Designed and facilitated a 10-week astronomy program for fourth and fifth grade students.
 - ❖ Created engaging, age-appropriate lesson plans covering a new astronomy topic each week, including planets, constellations, and the life cycle of stars.
 - ❖ Incorporated hands-on learning experiences, such as solar telescope viewing and evening stargazing sessions using a 22-inch Dobsonian telescope.
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AWARDED POSITIONS

Postbaccalaureate Research Assistant
(Position on hold due to 2025 NASA funding constraints)
NASA SURA CRESST II, Goddard Space Flight Center | Advisor: Dr. Avi Mandell

August 2025 – August 2026

- ❖ Selected for funded research under Dr. Mandell on exoplanet spectroscopy modeling with the Planetary Spectrum Generator.
- ❖ Work planned on an open-access exoplanet database (EMAC). Position paused due to funding cuts.

Accepted for Graduate Program

(Deferred due to national funding constraints)

Florida Tech University, Space Sciences Program

SKILLS

- ❖ Programming: Python (NumPy, SciPy, Matplotlib), Fortran (basic).
- ❖ Data Analysis: Spectroscopic reduction, EOS modeling, photometric extraction.
- ❖ Software: Microsoft Office, Google Suite.

MEMBERSHIPS

- ❖ Society of Physics Students at UC Santa Cruz
- ❖ Woman in Physics and Astrophysics at UC Santa Cruz
- ❖ American Astronomical Society
- ❖ Project Management Institute (PMI)

PUBLICATIONS

- ❖ **Larkins, L.**, Aguichine, A. 2026 Thermal effects on the bulk density of rocky planets: the Earth-like composition band. (in prep.)
- ❖ Aguichine A., Batalha, N., Nimmo, F., **Larkins, L.** 2026 Warm Bodies: partially molten rocky planets and the Earth-like radius band. (in prep.)

ORAL SCIENTIFIC CONTRIBUTIONS

- ❖ Contributed talk at the Other Worlds Laboratory Workshop July 2025
“Modeling Interiors of Molten Rocky Exoplanets”
Santa Cruz, California USA, 100 participants
- ❖ Science Communication Contributor - “Speak Up for Science” September 2025
Initiative, University of California
 - Invited to produce a short video highlighting the importance of scientific research and public support for science.
 - Featured on official University of California social media platforms, contributing to digital science communication and public engagement efforts.

OTHER COMMUNITY INVOLVEMENT

- ❖ Regular Attendee, UCSC Planetary Science Research Meetings Fall 2024 - Spring 2025